

Advanced Machine Learning

DATA 642

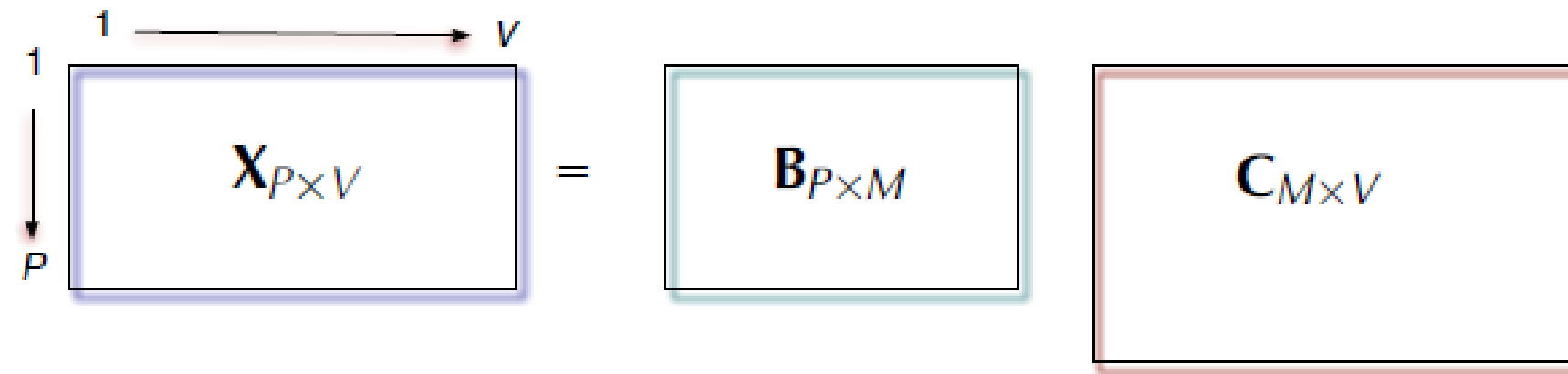
Lecture 7 — Introduction to Independent Vector Analysis

*Figures and notes are from the paper "Diversity in Independent Component and Vector Analyses: Identifiability, algorithms, and applications in medical imaging," (<https://ieeexplore.ieee.org/document/6784026>) and from the PhD thesis "Development of ICA and IVA Algorithms with Application to Medical Image Analysis," (<https://arxiv.org/abs/1801.08600>)

Latent variable methods

For matrix factorization, we

Let $\mathbf{X} = \mathbf{BC}$, where $\mathbf{C} \in \mathbb{R}^{M \times V}$ is the matrix that contains the underlying latent variables. We decompose the given set of observations $\mathbf{X} \in \mathbb{R}^{P \times V}$ as



e.g., using cost function

$$\min_{\mathbf{B}, \mathbf{C}} \|\mathbf{X} - \mathbf{BC}\|_p^2 + \lambda_1 \sum_{m=1}^M h(\mathbf{b}_m) + \lambda_2 \sum_{v=1}^V g(\mathbf{c}_v) \quad (1)$$

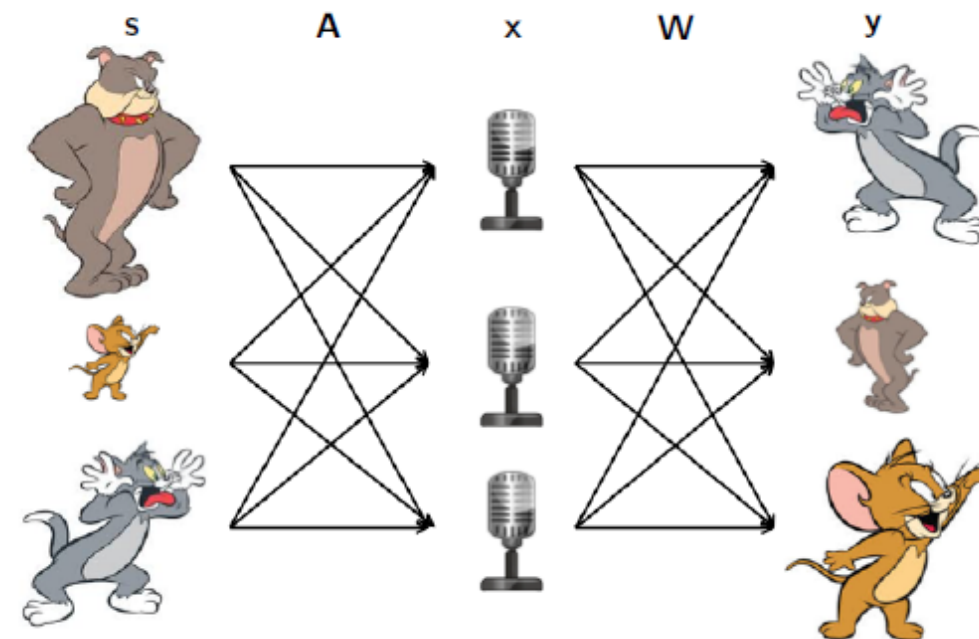
Independent component analysis (ICA)

Origins of ICA:

- ICA was developed in the 1980s and 1990s as a statistical technique aimed at solving the problem of separating a multivariate signal into components or sources. ICA was invented by Jeanny Hérault and Christian Jutten in 1985. (Wiki)

The Cocktail Party Problem:

- The cocktail party problem is a classic example used to explain ICA. It describes a situation where multiple people are talking simultaneously in a room, and the goal is to isolate each individual's speech from the mixed audio signals captured by multiple microphones.

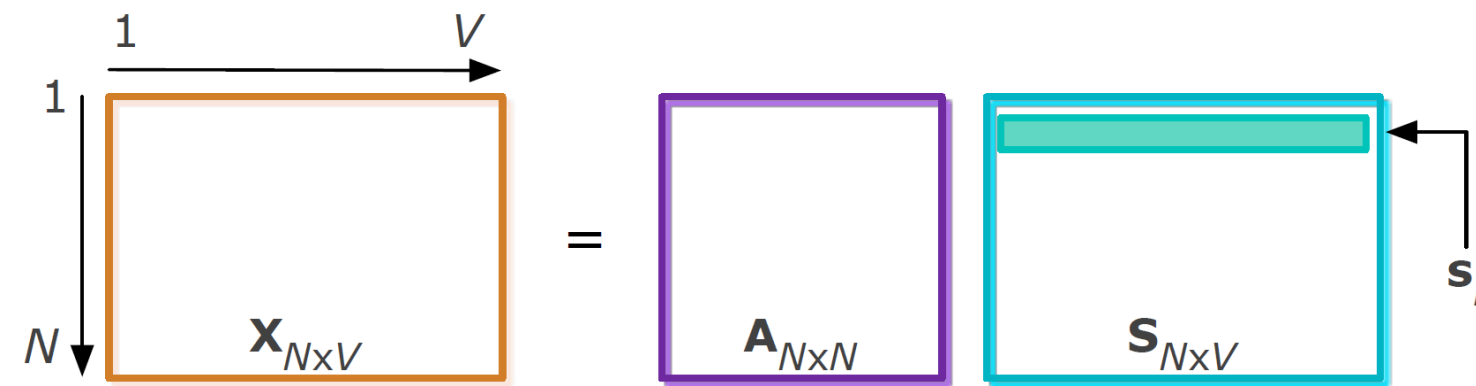


The model

The ICA model is given by

$$\mathbf{x} = \mathbf{A}\mathbf{s}, \quad (2)$$

where $\mathbf{x}$ are the observations and $\mathbf{s}$ are the latent sources linearly mixed by matrix $\mathbf{A}$.



ICA can uniquely identify the sources subject to scaling and permutation ambiguities by assuming **source statistical independence**.

The ICA objective function

An approach to define an ICA objective function is by the MI among the estimated sources and is defined by the Kullback Leibler (KL)-distance between the joint source density and the product of the marginal estimated source densities. Thus, by using the fact that $p_s(\mathbf{W}\mathbf{x}) = p_{\mathbf{x}}(\mathbf{x})|\det(\mathbf{W})|^{-1}$, the MI objective function is given by

$$\begin{aligned} J_{ICA}(\mathbf{W}) &= E \left\{ -\log \left[\frac{p_{s_1}(y_1)p_{s_2}(y_2) \cdots p_{s_N}(y_N)}{p_{s_1 s_2 \dots s_N}(y_1, y_2, \dots, y_N)} \right] \right\} \\ &= E \left\{ -\sum_{n=1}^N \log p_{s_n}(y_n) \right\} + E \{ \log p_s(\mathbf{y}) \} \\ &= \sum_{n=1}^N H(y_n) - H(\mathbf{y}) \\ &= \sum_{n=1}^N H(y_n) - \log |\det(\mathbf{W})| - H(\mathbf{x}), \end{aligned} \tag{3}$$

where the terms $H(y_n)$ and $H(\mathbf{x})$ are the (differential) entropy of the source estimates and the mixtures, respectively.

ICA algorithms

Solutions for the estimation of the components of the score function include parametric, non-parametric, and semi-parametric approaches.

- **Parametric Approaches:**

- **FastICA** , **efficient fast ICA (EFICA)**, and **information maximization (Infomax)** are based on a parametric approach.
- These methods use a fixed nonlinearity or model for the underlying distribution of the sources, making them computationally attractive.
- However, their separation performance suffers when the density of the true sources deviates significantly from the assumed model.

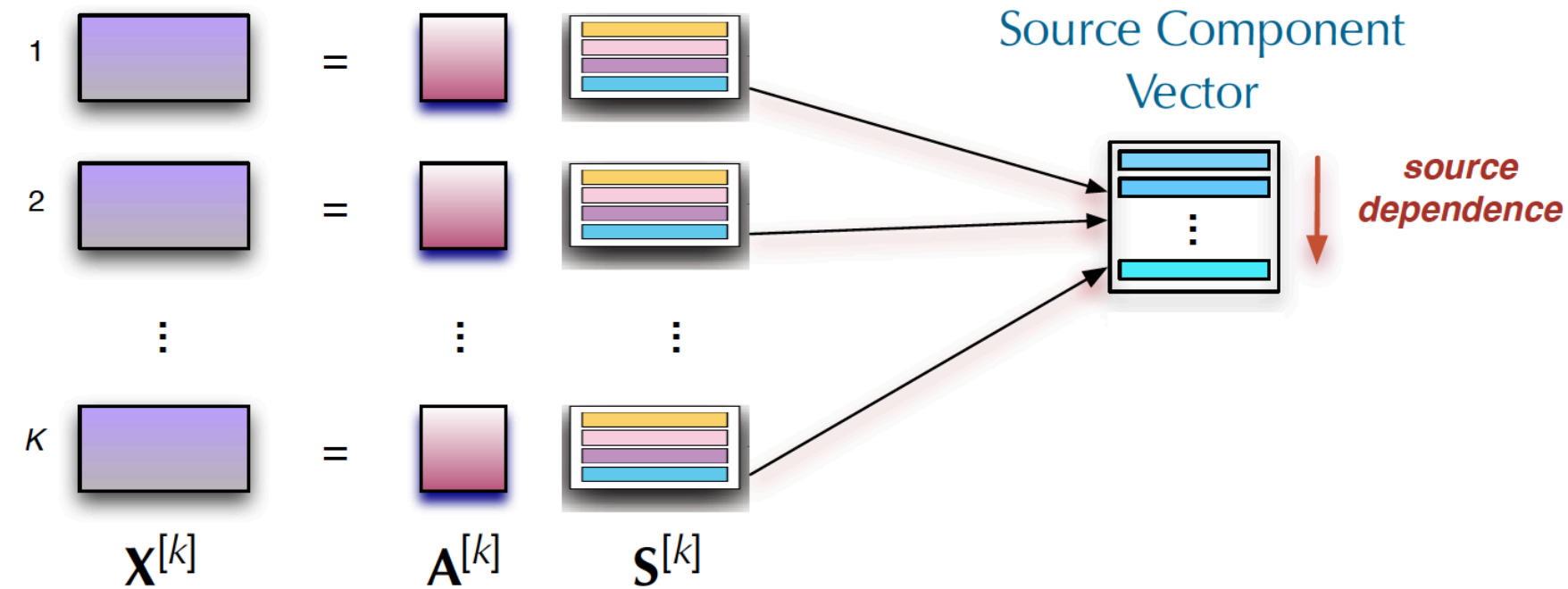
- **Non-Parametric Approaches:**

- **Robust, accurate, direct ICA (RADICAL)** is a nonparametric ICA algorithm, which uses estimates of entropy.
- Nonparametric methods are practically difficult due to the parameter selection required.
- They are also computationally demanding when the number of samples increases.

ICA algorithms (cont.)

- **Semi-Parametric Approaches:**
 - **ICA by entropy bound minimization (ICA-EBM)** and **Independent component analysis using semi-parametric density estimation via entropy maximization (ICA-EMK)** are based on semi-parametric approaches.
 - They provide flexible density matching through the use of measuring functions based on the maximum entropy principle.
 - These approaches can cover a wide range of PDFs but could be sometimes computationally expensive.

In many practical applications, multiple data sets with dependence among them need to be analyzed.



IVA is achieved, through the definition of the source component vector (SCV) that is defined by vertically concatenating the n th source from each of the K datasets and is denoted as

$$\mathbf{s}_n = [s_n^{[1]}, \dots, s_n^{[K]}]^\top, \quad (4)$$

where $\mathbf{s}_n$ is a K -dimensional random vector.

Goal: The goal in IVA is to estimate K demixing matrices to yield source estimates $\mathbf{y}^{[k]} = \mathbf{W}^{[k]} \mathbf{x}^{[k]}$, such that each SCV is maximally independent of all other SCVs.

The IVA problem is mathematically formulated similar to the ICA one, except that now we have K datasets $\mathbf{x}^{[k]}$, $k = 1, \dots, K$ where each dataset is a linear mixture of N statistically independent sources. Using random vector notation under the assumption that samples are i.i.d the noiseless IVA model is given by

$$\mathbf{x}^{[k]} = \mathbf{A}^{[k]} \mathbf{s}^{[k]}, \quad \mathbf{k} = \mathbf{1}, \dots, \mathbf{K}, \quad (5)$$

where $\mathbf{A}^{[k]} \in \mathbb{R}^{N \times N}$, $\mathbf{k} = \mathbf{1}, \dots, \mathbf{K}$ are invertible mixing matrices and $\mathbf{s}^{[k]} = [s_1^{[k]}, \dots, s_N^{[k]}]^\top$ is the vector of latent sources for the k th dataset. In the IVA model, the components within each $\mathbf{s}^{[k]}$ are assumed to be independent, while at the same time, the IVA model allows for dependence across corresponding components of $\mathbf{s}^{[k]}$ in multiple datasets.

The IVA objective function

Similar to ICA, IVA can be formulated in a ML framework. However, the optimization parameter is not just a single demixing matrix $\mathbf{W}$ as in the ICA case, but a set of demixing matrices $\mathbf{W}^{[1]}, \dots, \mathbf{W}^{[K]}$, which can be collected into a three dimensional array $\mathcal{W} \in \mathbb{R}^{N \times N \times K}$.

The MI objective function for the IVA model is given by

$$J_{IVA}(\mathcal{W}) = \sum_{n=1}^N H(\mathbf{y}_n) - \sum_{k=1}^K \log \left| \det \left(\mathbf{W}^{[k]} \right) \right| - H(\mathbf{x}^{[1]}, \dots, \mathbf{x}^{[K]}), \quad (6)$$

where $H(\mathbf{y}_n)$ denotes the entropy of the n th SCV and the term $H(\mathbf{x}^{[1]}, \dots, \mathbf{x}^{[K]})$ can be treated as a constant.

The gradient and optimization strategy

Using the IVA MI objective function the derivative w.r.t. to each of the demixing matrices is derived similar to the one for ICA and is given by

$$\frac{\partial J_{IVA}(\mathcal{W})}{\partial \mathbf{W}^{[k]}} = E \left\{ \boldsymbol{\phi}^{[k]} (\mathbf{x}^{[k]})^\top \right\} - (\mathbf{W}^{[k]})^{-\top}, \quad (7)$$

where $\boldsymbol{\phi}^{[k]} = - \left[\frac{\partial \log p_{s_1}(y_1)}{\partial y_1^{[k]}}, \dots, \frac{\partial \log p_{s_N}(y_N)}{\partial y_N^{[k]}} \right]^\top$. Thus, each of the K demixing matrices is updated using

$$(\mathbf{W}^{[k]})^{\text{new}} \leftarrow (\mathbf{W}^{[k]})^{\text{old}} - \gamma \frac{\partial J_{IVA}(\mathcal{W})}{\partial \mathbf{W}^{[k]}}, \quad (8)$$

where γ is the step size.

IVA algorithms

In contrast to ICA, most IVA algorithms developed to date are based on parametric methods for the estimation of the score function during adaptation.

- **IVA-Laplacian (IVA-L):**

- Takes higher-order statistics (HOS) into account and assumes a Laplacian distribution for the underlying source component vectors.

- **IVA-Gaussian (IVA-G):**

- Exploits linear dependencies but does not take HOS into account.
- Assumes a Gaussian distribution for the underlying sources, simplifying the gradient of the MI objective function and making the Hessian positive definite.
- Enables practical use of second-order algorithms such as Newton-variants, improving convergence quality.

IVA algorithms (cont.)

- **IVA-Generalized Gaussian (IVA-GGD) and (IVA-A-GGD):**
 - A more general IVA implementation considering both second and higher-order statistics.
 - Assumes a multivariate generalized Gaussian distribution (MGGD) for the underlying sources.
 - Multivariate Gaussian and Laplacian distributions become special cases through parameter estimation.
- **IVA by multivariate entropy maximization with kernels (IVA-M-EMK):**
 - Can be perfectly applied in problems where the knowledge of the multidimensional PDF that best matches the underlying properties of the data is necessary.
 - In certain scenarios as K and N increase algorithm becomes computationally attractive.